Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20251103-041-R0	Issued	2025-11-03

UT Re-measurement - 2-RF-102

Scope and Responsibilities

This report details the ultrasonic thickness re-measurement of line 2-RF-102 within the ATU-100 at the Hamilton Plant. The scope is limited to the 12-month follow-up of specific Condition Monitoring Locations (CMLs) identified as being below the structural limit in previous surveys. The inspection was performed to verify current corrosion rates and validate the remaining life projections prior to the scheduled turnaround.

Sarah Thompson, acting as the UT technician, was responsible for the field data acquisition and preliminary analysis. David Park, serving as the Quality Manager, provided the technical review and oversight of the findings. All activities were conducted in accordance with the site-specific requirements for the Amine Treating Unit to ensure asset integrity and compliance with corporate safety standards.

Inspection Methodology and Calibration

Ultrasonic thickness measurements were performed under QSCP-NDT-001 Rev 4. This procedure incorporates the requirements of ASTM E797 and ASNT SNT-TC-1A for the qualification of NDE personnel. The inspection utilized an Olympus 38DL PLUS 2901456, which was verified against LNDE-CAL-2025-0961 prior to the commencement of work on 2-RF-102. The gauge was calibrated using the stainless-steel velocity setting appropriate for the A312 TP316 material.

The survey focused on the downstream sections of the reflux flow-control valve where accelerated thinning was previously noted. Data acquisition followed the guidelines set forth in API 570 and ASME B31.3 to ensure the structural integrity of the piping system. The measurement process included CML-id validation at the point of data export as required by the updated site NDE protocols.

Historical Context and Reference Data

The requirement for this 12-month re-measurement was established in QSCP-IP-ATU100 Rev 4 following the results of the five-year survey documented in LNDE-RPT-20241025-031-R2. That survey identified significant wall loss in the reflux circuit that deviated from the near-nil corrosion observed in 3-RF-101. The specific locations were formally flagged for monitoring to track the degradation rate against the projected turnaround date.

The current inspection also serves to address the status of QSCP-NCR-2024-032, which was issued when four CMLs fell below the t-min of 2.16 mm. The nonconformance report noted that the thinning was localized to the downstream side of the control valve, with a recorded loss of approximately 0.9 mm compared to the 2019 baseline data. This re-measurement confirms if the corrosion rate has stabilized or accelerated.

Findings and Integrity Assessment

The re-measurement of 2-RF-102 confirmed that wall thicknesses at the critical locations remain below the retirement limit of 2.16 mm. Recorded values are as follows: CML 2-RF-102-04 at 1.6 mm, 2-RF-102-05 at 1.69 mm, 2-RF-102-06 at 1.67 mm, and 2-RF-102-07 at 1.75 mm. Based on these readings, the remaining life for these points is considered expired according to the site rules for Class 2 reflux service.

Analysis of the data indicates that the corrosion rate remains unchanged since the previous assessment. Although the readings are below the structural t-min, the projected degradation does not exceed the safety margins required before the upcoming maintenance window. Consequently, the recommendation for replacement during the August 2026 turnaround, as specified in the piping inspection plan, remains valid and unchanged.

Thickness readings, measured 2025-11-03

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2024-11-04	3.55	3.52	0.013	0.023	>50	2031-11-03	
2-RF-102-02	3.91	2.16	2024-11-04	3.53	3.49	0.013	0.026	>50	2031-11-03	
2-RF-102-03	3.91	2.16	2024-11-04	3.65	3.64	0.010	0.007	>50	2031-11-03	
2-RF-102-04	3.91	2.16	2024-11-04	1.94	1.60	0.077	0.203	expired	2026-11-03	< t-min; gauge photo attached

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-05	3.91	2.16	2024-11-04	1.92	1.69	0.165	0.197	expired	2026-11-03	< t-min; gauge photo attached
2-RF-102-06	3.91	2.16	2024-11-04	1.92	1.67	0.168	0.177	expired	2026-11-03	< t-min; gauge photo attached
2-RF-102-07	3.91	2.16	2024-11-04	1.86	1.75	0.075	0.177	expired	2026-11-03	< t-min; gauge photo attached

Sign-off	Name	Function	Date	Signature
Prepared	Sarah Thompson	UT technician, SNT-TC-1A UT Level II	2025-11-03	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2025-11-03	_____